

SignSafe AI: Agreement Intelligence Platform Using Retrieval-Augmented Generation (RAG)

Problem Statement

In today's digital world, users frequently accept agreements, contracts, internship offer letters, rental agreements, service policies, and Terms & Conditions without fully understanding their contents. These documents are often lengthy and written in complex legal language, making it difficult for individuals to identify important clauses, obligations, penalties, renewal conditions, and potential risks before signing.

As a result, users may unknowingly agree to unfavorable terms such as automatic renewals, hidden penalties, restrictive confidentiality clauses, excessive liabilities, or data-sharing permissions.

To address this challenge, the proposed system, **SignSafe AI**, is an AI-powered Agreement Intelligence Platform that helps users understand agreements before acceptance. The system allows users to upload agreement documents in PDF format and automatically analyzes the content using Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs).

The platform performs intelligent document analysis by generating concise summaries, extracting key clauses, identifying potential risks, and enabling users to ask questions about the uploaded agreement through a conversational interface. By combining semantic search using FAISS vector databases with Google's Gemini AI model, the system provides context-aware and document-grounded responses, reducing misinformation and improving transparency.

Objectives

- Simplify complex legal and contractual documents.
- Generate concise agreement summaries.
- Extract important clauses such as Payment, Termination, Renewal, Confidentiality, and Liability.
- Identify potential risks and obligations within agreements.
- Enable users to interact with agreements through a RAG-powered AI assistant.
- Improve user awareness and decision-making before signing agreements.

Expected Outcome

The proposed system provides a user-friendly platform that enhances agreement comprehension, highlights critical information, and assists users in making informed decisions by leveraging Artificial Intelligence, Natural Language Processing, and Retrieval-Augmented Generation techniques.